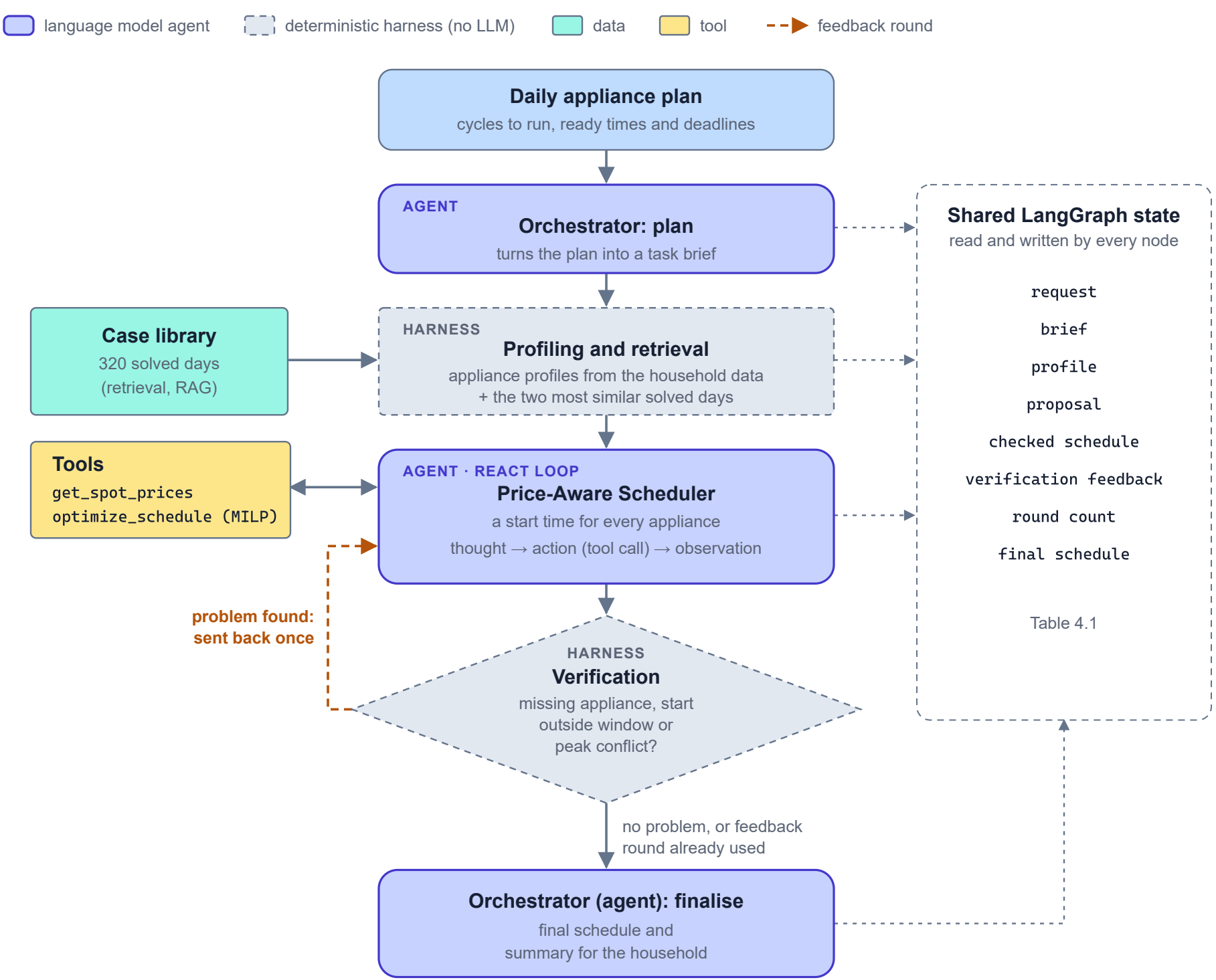

language model agent
deterministic harness (no LLM)
data
tool
feedback round
Daily appliance plan
cycles to run, ready times and deadlines
AGENT
Orchestrator: plan
turns the plan into a task brief
Case library
320 solved days
(retrieval, RAG)
HARNESS
Profiling and retrieval
appliance profiles from the household data
+ the two most similar solved days
Tools
get_spot_prices
optimize_schedule (MILP)
AGENT · REACT LOOP
Price-Aware Scheduler
a start time for every appliance
thought → action (tool call) → observation
problem found:
sent back once
HARNESS
Verification
missing appliance, start
outside window or
peak conflict?
no problem, or feedback
round already used
Orchestrator (agent): finalise
final schedule and
summary for the household
Shared LangGraph state
read and written by every node
request
brief
profile
proposal
checked schedule
verification feedback
round count
final schedule
Table 4.1